Reinstatement out of equilibrium: shock speed and the destination of new work

Joakim Storck

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Abstract

This paper turns a spatial task-based model of automation into laws of motion and asks how the outcome of a technological transition depends on the speed of the shock. New tasks arrive as a flow into a stock of unbound work; occupations bind the stock through a match-driven claim at a rate limited by their size. One ratio governs the outcome, the tempo of the shock against the tempo of absorption. Both endpoints are proved as limits of the binding law, for every capacity-scaling exponent, and the crossover is derived in closed form in a two-occupation economy. Between the limits the destination moves monotonically with the tempo: a gradual shock matches the reinstated work to well-suited specialists, a congested one forces it onto large, lower-paid occupations, and the two allocations correlate at +0.37 under the linear reference capacity law (+0.68 weighted by employment). A general-equilibrium price close is stabilising and raises the labour share. The realisation is a calibrated illustration and carries no causal claim.

1 Introduction

A technological breakthrough can happen overnight. Its consequences take years to unfold, and while they do, the labour market is already responding. Firms invest, new work practices spread, training programmes equip engineers and workers with the skills the technology demands, and schools carry the new knowledge to the next cohort. As the technology diffuses and matures it displaces old tasks and introduces new ones. Some workers find new work where they stand, others move to reach it. These processes run in parallel, and each runs at its own speed.

Research has measured these speeds separately. The diffusion literature has timed the technology since Griliches (1957) fitted logistic curves to the spread of hybrid corn. New technologies take decades to reach their ceilings, general-purpose technologies deliver their gains a generation after they arrive, and adoption lags differ widely across countries (David 1990; Comin and Hobijn 2010). A second clock, less often read as one, times the occupations. Task content turns over inside narrowly defined job titles across decades (Atalay et al. 2020), new titles appear where adaptation to new knowledge is under way (Lin 2011; D. Autor et al. 2024), and hiring a skilled worker costs a firm ten to seventeen weeks of wages with a marginal cost that rises in the number of hires (Blatter, Muehlemann, and Schenker 2012), so an occupation takes on new work at a finite, size-dependent pace. A third clock times the workers. Occupational moves are slow and costly because much of human capital is specific to tasks, and a mover loses the part that does not transfer (Kambourov and Manovskii 2009; Gathmann and Schönberg 2010). The task-based tradition supplies the content of the shock. Automation displaces labour from existing tasks, the creation of new tasks reinstates it, and the balance between the two sets the demand for labour (Acemoglu and Restrepo 2018; Acemoglu and Restrepo 2019). Each clock has been measured on its own. No framework runs them against each other and asks who ends up with the new work.

The task-based framework reads the consequences of a technology as the difference between two rest points. Its dynamics, where it has them, belong to the technology, to which tasks are automated and when new ones arrive, while workers reach their new allocation instantaneously (Acemoglu and Restrepo 2019). The intermediate path of the allocation drops out, and with it the cost of the transition and any dependence of the outcome on the speed of the shock. Yet the transition is where the human consequences of automation are felt. The central claim of this paper is that the speed of the shock relative to the speed of adjustment determines not merely how costly the transition is but *who ends up doing the reinstated work.* The adjustment speed that matters turns out to be the occupations', not the workers'. Both frictions run in the model, and the decomposition of Section 4.2 separates them: slowing worker mobility fifteen-fold leaves the destination essentially unchanged, while slowing absorption alone reproduces the downgrading in full.

Worker mobility, the margin the adjustment literature has measured most closely, is the friction that turns out not to bind.

The framework we build on is developed in two papers. Storck and Andersson (2026) map tasks and occupations from O*NET statements onto a two-dimensional disk, where each occupation is a bundle of tasks. Storck (2026) builds the model of technological change on that geometry. The market price of skill varies over the space as a directional field, and a technology arrives as a Gaussian augmentation field. Automation removes the high-priced tasks of a bundle. Reinstatement seeds new tasks on the gradient boundary of the field, and they survive as human work where labour remains the cheaper producer. That model is closed as a static equilibrium, with the adjustment dynamics left explicitly for later work. The rise-and-fall of employment which Bessen (2019) documents over time appears in it only as a cross-section over the demand elasticity, not as a trajectory. This paper is that later work.

We make one structural addition and otherwise reuse the static model unchanged. That model creates new task mass on the boundary ring and lets it meet three fates. Capital captures one part and occupations bind another, while the unbound remainder is created and discarded at every evaluation, with no home in the model. The dynamic model gives it a home, a stock $U(\mathbf{r}, t)$ of unbound task mass that fills as the technology matures and drains as occupations bind it. This stock is the only genuinely new object; population, price, displacement, seeding, and the bundle operator are the static model's, now read as rates rather than as a solved equilibrium. The trajectory of the stock, $U(t)$, is the transitional mismatch between destruction and creation, and it is the first observable the static model cannot produce.

The second, and the paper's central result, concerns binding. Unbound work seeks the occupations whose held capabilities best match it, but an occupation can take on new work only at a rate limited by its size, so a small best match saturates and leaves a residue for the next-best match. The shock seeds new work at one tempo and occupations bind it at another, so the realised allocation depends on a race. We show that the ratio of the two tempos governs not only the height and duration of the transitional mismatch $U(t)$ but the final destination of the work. A gradual shock lets the best-matched occupations keep pace, and the high-skill specialists whose capabilities fit the frontier bind the reinstated work. A shock that outruns redistribution forces the work onto the large, lower-skilled occupations that can absorb it fastest, a downgrading that the gradual transition avoids. The two regimes run through the same technology and the same field, and their final allocations of reinstated work correlate at $+0.37$ under the linear reference capacity law, unweighted, and $+0.68$ weighted by employment. Both endpoints are proved as limits of the binding law: absorption faster than the shock converges to the match-weighted allocation (Proposition 1), absorption slower than the shock to an allocation in the initial masses alone, for every capacity-scaling exponent (Proposition 2), and a two-occupation economy locates the crossover between the limits in closed form (Proposition 3). The realised destination moves monotonically between the limits as the tempo varies (Section 4.3). A general-equilibrium close endogenises the price level; the feedback is stabilising, and the labour share rises relative to fixed prices (Section 5). The destination itself is path-dependent, and the equilibrium framework sees none of it. The contribution is the binding law and these limit theorems; the calibrated economy they run through is an illustration in the discipline of Storck (2026) and carries no causal claim.

That reallocation takes time and shapes equilibrium is an old observation. In the islands tradition workers move slowly between markets and unemployment is the reallocation in progress (Lucas and Prescott 1974; Alvarez and Shimer 2011); distance appears as a cost, but there is no space in which the shock and the workers' capabilities are located together. Şahin et al. (2014) measure the unemployment produced when job seekers search in the wrong markets: workers waiting for work. Guvenen et al. (2020) measure mismatch inside the match, as the distance between the skills an occupation requires and those its worker brings. The unbound stock of this paper is the reverse object, work that has been created but not yet attached to any occupation, mismatch upstream of any vacancy. Its hump is the model's analogue of their cyclical index, driven by task destruction and creation rather than by the business cycle. Closest on the question of tempo, Adão, Beraja, and Pandalai-Nayar (2024) relate the speed of a technological transition to skill specificity. When the benefited activities require skills far from the rest of the economy, incumbents cannot reallocate and demand waits for the entry of new generations. In their model the economy converges to the endpoint the technology pins down, regardless of the path. Here the tempo of the shock races a fixed speed of reallocation, and the endpoint moves with it. The same technology, run slowly or quickly through the same economy, binds the reinstated work to different occupations. The tempo is also a policy margin. Guerreiro, Rebelo, and Teles (2022) find it optimal to tax robots while the workers who cannot move remain in the labour force, and Beraja and Zorzi (2025) to slow automation while displaced workers reallocate under borrowing constraints. Our result gives that lever a second consequence: pacing moves the destination of the work.

2 The static foundation

We recall only what the dynamics require. Storck and Andersson (2026) construct and validate the geometry, the disk, its coordinates, and the mapping of tasks and occupations into it. Storck (2026) builds the model of technological change on that geometry and supplies the calibration used here.

Tasks lie on the unit disk in polar coordinates $\mathbf{r} = (\xi, \chi)$, with ξ a domain orientation and χ a depth. The market price of skill is a directional field

$$\ln \Pi(\mathbf{r}) = m_0 + m_1 \cos\xi + m_2 \sin\xi + \chi\,(m_3 + m_4 \cos\xi + m_5 \sin\xi), \tag{1}$$

high in the socio-cognitive directions and low in the service and manual ones: the wage structure of Storck and Andersson (2026) read spatially, lifted from occupational centroids to task locations by Storck (2026). Estimated on the occupational wage cross-section ($N = 785$; HC3 standard errors in parentheses), the field is

$$\ln \Pi(\mathbf{r}) = \underset{(0.036)}{3.420} + \underset{(0.048)}{0.192}\,\cos\xi + \underset{(0.051)}{0.089}\,\sin\xi + \chi\Big(\underset{(0.090)}{-0.065} + \underset{(0.112)}{0.008}\,\cos\xi + \underset{(0.134)}{0.891}\,\sin\xi\Big), \tag{2}$$

with $R^2 = 0.523$; the well-paid professional and managerial occupations sit in the socio-cognitive directions of the disk. Two structural checks discipline the form, an isotropic depth term indistinguishable from zero and a single first harmonic for the depth return; Appendix A carries them with the estimation detail and the provenance of every inherited parameter. Depth is what Adão, Beraja, and Pandalai-Nayar (2024) call skill specificity. The geometry gives it a direction, and the price field decides where specificity pays. A technology is a Gaussian field $\phi_K(\mathbf{r}) = A_K \exp(-\frac{1}{2}\|\mathbf{r} - \mathbf{p}_K\|^2/z_K^2)$, with maturity A_K, centre $\mathbf{p}_K$, and reach z_K. Capital bears a task only where it is the cheaper producer, so the operated share is

$$a(\mathbf{r}) = \Big[1 + \exp\!\Big(-\big(s_K\phi_K(\mathbf{r}) - R/\Pi(\mathbf{r})\big)/\tau\Big)\Big]^{-1}, \tag{3}$$

where τ is the width of the adoption margin. The condition crosses first where the price Π is high, so automation takes the dearest work. Each occupation o is a bundle $b_o(\mathbf{r})$ with centroid μ_o; its displaced mass is $D_o = \int b_o a$, and the aggregate displacement is $\Delta\Gamma^D = \sum_o L_o D_o$.

Reinstatement creates new task mass $M = \gamma\,\Delta\Gamma^D$ on the gradient ring of the field, $\hat{g}(\mathbf{r}) = \|\nabla\phi_K\|/\int\|\nabla\phi_K\|$, which peaks at distance z_K from the centre. A seeded task survives as human work only where the operated share is low, through the survival gate $(1-a)$: deep in the core, where capital dominates, the same seed is captured by capital. The surviving seed density is therefore $\hat{g}\,(1-a)$, concentrated on the boundary crescent. Figure 1 draws this inherited economy from the pipeline: the price contours, the occupation centroids, and the technology field whose gradient ring carries the seeding.

This construction generalises the reinstatement of Acemoglu and Restrepo (2018), and the geometry of their model motivates ours. Production there combines a unit measure of tasks on an interval. Automation raises a lower threshold below which tasks can be produced by capital, and new tasks arrive at the upper endpoint, so reinstatement is boundary creation already on the line. Whether an automatable task is actually produced by capital is a cost comparison, capital where the rental rate beats the effective wage,

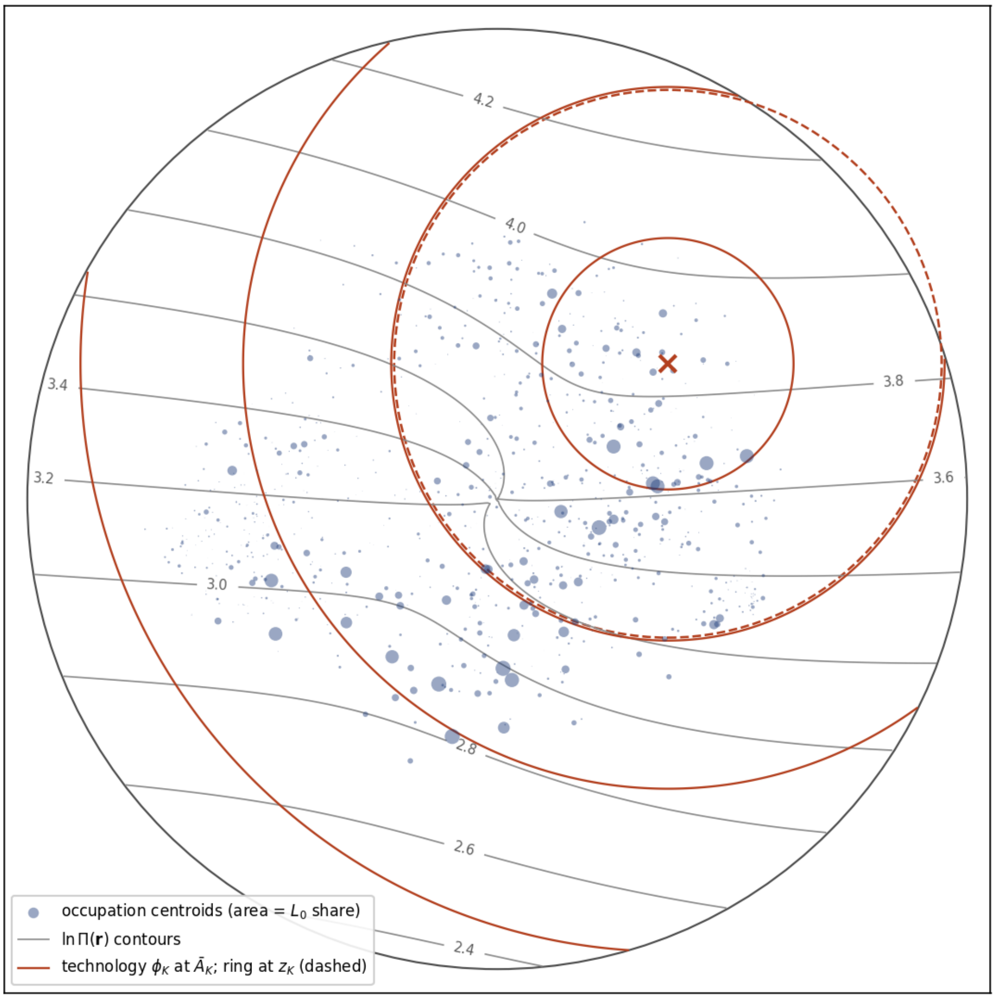

Figure 1: The inherited economy. Grey: contours of the estimated log price field, eq. (2). Blue: occupation centroids, area proportional to pre-shock employment share L_0. Red: the calibrated technology field ϕ_K at mature amplitude, its centre marked and the gradient ring at radius z_K dashed; the surviving seed concentrates where the ring meets the low-priced side of the adoption boundary. Generated from the same objects the dynamic scripts consume.

and equation (3) makes the same comparison pointwise on the disk. Two features of their boundary are assumptions that the disk turns into outcomes. Labour's comparative advantage grows with the task index, so the new task at the endpoint is the one where labour's advantage is greatest, and a bound on the capital stock guarantees that labour adopts it immediately (Acemoglu and Restrepo 2018). Here the endpoint becomes the gradient ring, the ordering becomes the price field, and the guarantee becomes the gate, which prices survival region by region. The new tasks in which capital holds the advantage, noted and set aside in Acemoglu and Restrepo (2019), are the captured fate $\zeta \cdot a$. The generalisation is what opens the destination question. On the line new work has one place to appear. On the ring it appears in every direction around the field, and which occupations end up holding it becomes an economic outcome rather than a location fixed by construction.

An occupation can take a seeded task only where its capabilities match the task's requirements. The implemented readiness is symmetric and local,

$$e_o(\mathbf{r}) = \exp\Big[-\Big(\delta_o(\mathbf{r}) + \lambda_{\text{over}}\,\bar{\delta}_o(\mathbf{r})\Big)/\ell\Big]\ \exp\Big(-\|\mathbf{r} - \mu_o\|/\rho\Big), \tag{4}$$

where $\delta_o = \sum_k v_k \max(q_k(\mathbf{r}) - q_{o,k}, 0)$ is the v-weighted capability deficit, $\bar{\delta}_o$ the corresponding surplus, ℓ the acquisition scale, ρ the attachment locality, and λ_{over} the over-qualification weight. Each parameter has a stated derivation. The requirement fields q_k are capability planes over the disk, restated in Appendix A. The weights $v = (0.328, 0.049)$ on the priced clusters $K = \{S1, S2\}$ are wage returns from a replication of the mediation regression of Storck and Andersson (2026), and the scale $\ell = 0.133$ is set by the standard-deviation rule of Appendix A. The attachment capacity of a location is $C(\mathbf{r}) = \sum_o L_o e_o(\mathbf{r})$. This primitive is shared verbatim between the two papers and between the static and dynamic layers of the implementation.

All parameters of this foundation are inherited unchanged from the economy table of Storck (2026): $R = 18$, $\tau = 0.08$, $\beta = 0.5$ (congestion), $\gamma = 0.5$, $\ell = 0.133$, $\rho = 0.5$, $\lambda_{\text{over}} = 1$. The mobility scale ν (one standard deviation of baseline occupation value per logit unit) and the move cost c_m (the median move costs one ν) follow the same rules, evaluated here at the pre-shock baseline $A_K = 0$, the state the dynamics start from, giving $\nu = 11.8$ and $c_m = 23.0$. That table evaluates the same rules at the calibrated technology and reports 11.6 and 22.6. The two per cent difference moves no result at its reported precision. Table 3 in Appendix A consolidates every parameter of the economy with its provenance: estimated, calibrated, rule-derived, normalised, or swept.

Two features of this foundation are load-bearing for what follows. First, displacement falls *within* occupations, on the interior margin $a(\mathbf{r})$, not between regions. Second, the survival gate places the surviving reinstatement on the low-priced side of the boundary. Humans retain the work that is not worth automating, which in this price field is the cheaper, lower-Π work. The destination results below interact with this second feature, and we return to it in the discussion.

3 From equilibrium to motion

The static model computes a rest point $L = F(L)$. The present model attaches calendar time to the same primitives and studies the path of the allocation while technology matures. Technology maturity, population, unbound task mass, and occupational task content become state variables. Displacement, seeding, binding, and reallocation become transition rates. The task geometry supplies the constitutive laws for these rates. Prices set adoption, the boundary field locates reinstatement, readiness ranks viable destinations, and occupation size limits absorption. The resulting state accounting gives the transition an observable stock, $U(\mathbf{r}, t)$, which records new human work before any occupation has taken it on.

3.1 Transition accounting and the unbound stock

In the static model the seeded mass M meets three fates. Capital captures a part $\zeta \cdot a$, occupations bind a part $\zeta \cdot (1-a)\,\Phi$, and a part $\zeta \cdot (1-a)\,(1-\Phi)$ remains unbound, where $\zeta = M\hat{g}$ is the seed density and Φ is the bound share; the three parts sum to ζ. The unbound part performs no work and does not enter the density; it is created and discarded at each evaluation. The dynamic model gives it a stock,

$$\partial_t U(\mathbf{r}, t) = \dot{\zeta}(\mathbf{r}, t) - \sum_o \iota_o(\mathbf{r}, t), \qquad \dot{\zeta}(\mathbf{r}, t) = \gamma\,[\partial_t\,\Delta\Gamma^D]_+\,\hat{g}(\mathbf{r})\,(1 - a(\mathbf{r}, t)), \tag{5}$$

where $\dot{\zeta}$ is the surviving seeding rate and ι_o the binding flow to occupation o (§3.3). The seeding follows the *rate* of displacement. New tasks enter while the technology matures. Seeding ceases once displacement plateaus. The unbound stock is the sole structural addition; the remaining objects are the static model's primitives read as rates.

Figure 2 gives the transition accounting as a stock-and-flow diagram in the notation of Forrester (1961). Automation creates a timed, survival-gated inflow of new task mass. The binding rate drains $U(\mathbf{r}, t)$ into occupation-bound work, and its information inputs are the law of Section 3.3: the match readiness that allocates the flow, and the size margin that limits its rate, fed back by the bound mass itself while the saturated residue stays in the stock. That the match readiness enters as a constant is the frozen-readiness assumption of Section 3.3 in symbol form. The population is a level of its own, moved by a personnel flow that follows value; no line runs from the population into the binding chain, which is the decomposition result of Section 4.2 drawn as structure.

Task mass enters through seeding and leaves U through binding. Population is

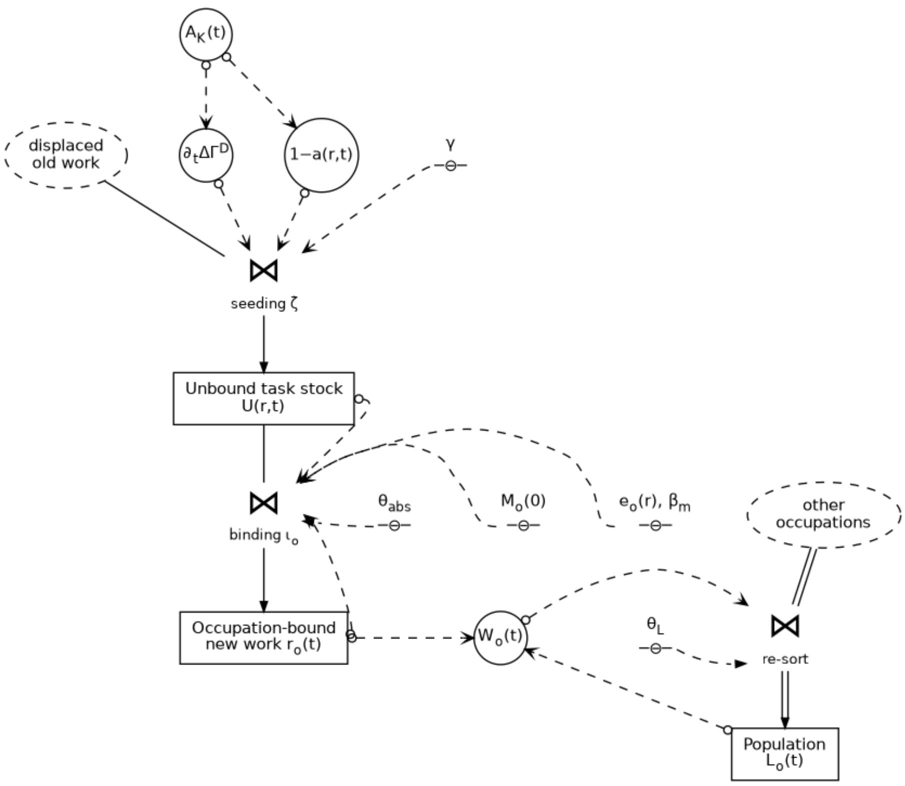

Figure 2: Transition accounting for displaced and reinstated work, in the notation of Forrester (1961): rectangles are levels, valves rates in the flow channels, circles auxiliary variables, a name over the bar-and-circle symbol a parameter, and dashed ellipses sources outside the accounting. Solid lines carry task mass, the double line the personnel flow, and dashed lines information, with a take-off circle at the source. The seeding rate is eq. (5), the binding rate is governed by eqs. (8) and (9), and the re-sort by eq. (6). The figure is generated from the implementation, with every edge audited against a code witness.

conserved, $\sum_o \dot{L}_o = 0$, and the created task mass is assigned among three fates: capital capture, occupation binding, and unbound carry-over. The stock U closes this ledger by giving the carry-over term a state variable.

3.2 Population and maturation

The population moves toward the static reallocation target, but from the *current* distribution rather than a frozen baseline:

$$\dot{L}_o = \tfrac{1}{\theta_L}\Big(T_o(L,t) - L_o\Big), \qquad T_o(L,t) = \sum_{o'} L_{o'}(t)\, P\Big(o \mid o';\, W(L,t)\Big), \tag{6}$$

with P a row-stochastic origin–destination logit, so $\sum_o \dot{L}_o = 0$ exactly. The timescale θ_L is a calibrated speed in interpretable units. It replaces the static solver's damping and inherits nothing from it. First order is deliberate because it yields lags and gaps but no overshoot.

The technology matures as a logistic diffusion anchored in calendar time, the S-curve that has described technology adoption since Griliches (1957). With t in years and a shock window T_{shock}, the slope is set so that A_K rises from five to ninety-five per cent of its mature value over T_{shock} years, centred at $T_{\text{shock}}/2$:

$$A_K(t) = \frac{\bar{A}_K}{1 + \exp\Big(-(5.88/T_{\text{shock}})\,(t - T_{\text{shock}}/2)\Big)}. \tag{7}$$

Anchoring the shock in calendar time makes θ_L and the binding timescale below calendar quantities, so that the outcome is governed by the *ratio* of the shock tempo T_{shock} to the redistribution tempo. We set $T_{\text{shock}} = 5$ years throughout. The value is a normalisation of the time axis, not a calibration: every rate in the model scales as $1/\theta_L$ or $1/\theta_{\text{abs}}$ or is set by the window, so only the ratio enters the results, up to the numerical step, and the sweep over the redistribution tempo at fixed T_{shock} traverses it.

3.3 Binding: match allocates, size limits the rate

The binding flow ι_o rests on a separation that proved load-bearing. Match sets *where* the unbound mass goes, and size sets *how fast* it binds. At each location the mass is attracted to the best match through a sharpened share, and each occupation binds up to a size-limited capacity:

$$t_o(\mathbf{r}) = U(\mathbf{r})\, \frac{e_o(\mathbf{r})^{\beta_m}}{\sum_{o'} e_{o'}(\mathbf{r})^{\beta_m}}, \tag{8}$$

$$c_o = M_o/\theta_{\text{abs}}, \quad M_o = M_o(0) + r_o, \qquad \iota_o(\mathbf{r}) = t_o(\mathbf{r})\, \min\Big(1,\, \frac{c_o}{\int t_o\, d\mathbf{r}}\Big). \tag{9}$$

The claim t_o distributes the unbound mass by sharpened match. The readiness e_o is the shared attachment primitive of eq. (4), and the exponent $\beta_m > 1$ lets the best match dominate the claim at each location. The capacity c_o scales with the occupation's task mass M_o, which grows endogenously with the reinstated mass r_o it has already bound. What a saturated occupation cannot staff this step, the fraction $1 - \min(1, c_o / \int t_o)$, returns to U and is offered at the next step to the next-best match. Large occupations therefore bind faster in absolute terms but not faster relative to their size, and the residue cascades down the match ranking.

The same primitive sets the static attachment capacity C and allocates the unbound mass here; the static and dynamic layers of the implementation share it exactly. Its symmetry carries the economics. Being able to do a task and becoming its home are different questions, and penalising over-qualification alongside the deficit keeps an occupation far above a task's level from drawing the task in as its own work. Storck (2026) states the form and its one-sided limit ($\lambda_{\text{over}} = 0$, $\rho \to \infty$). What the dynamic layer adds is only the sharpening β_m and the size-limited rate.

The form was chosen against an alternative that conflates the two roles. A binding rate proportional to size times match, $\iota_o \propto M_o\, e_o$, lets size *multiply* match. In that form the absolute absorption correlates with occupation size at +0.99 and with the unconstrained claim at +0.07 at the reference tempo, so size decides the destination and match governs only the relative growth. The match-allocated form with a size cap inverts this. Absorption tracks the claim at +0.73, and the size correlation falls to +0.13 at the reference tempo and to +0.04 in the gradual regime, with size demoted from setting *where* the work goes to *how fast* it binds, which is its proper role. The sharpness β_m and the absorption timescale θ_{abs} are calibration handles rather than worldviews, and their economic content, how sharply the best match wins and whether absorptive capacity scales linearly or sub-linearly with size, is settled by the sweeps of Section 4.2. Their economic interpretation is direct. The sharpness β_m sets how strongly the best match dominates the local claim: $\beta_m = 1$ shares the mass in proportion to fit, $\beta_m \to \infty$ assigns each location strictly to its best match; its calibration target is the observed concentration of new work across occupations. The absorption timescale θ_{abs} is the time an occupation needs to absorb its own task mass in new work, an onboarding capacity per unit size; its target is the observed pace at which occupations take on new task content. The runs fix $\beta_m = 3$ and sweep the tempo; Section 4.2 reports the result's sensitivity to β_m over 1 to 8 and shows that the destination result is carried by θ_{abs} alone.

Three structural assumptions. The path-dependence of the results rests on three assumptions of this law, stated here together. *Irreversible binding*: bound mass never returns to the unbound stock, so early allocations are not revised later; partial release and task decay are extensions noted in Section 6. *Frozen readiness*: the fields e_o are time-invariant absent births. This is a timescale separation, capability acquisition slower than both the shock and the absorption, and it is the mechanism of Adão, Beraja, and Pandalai-Nayar (2024), where incumbents cannot acquire the frontier skill within the transition and adjustment waits for entry. The assumption is released and tested in Section 4.2: the update variant widens the divergence, so the frozen baseline is the conservative side of the ambiguity. *Capacity scaling*: the linear law $c_o = M_o/\theta_{\text{abs}}$ of eq. (9). Proposition 2 is proved for every exponent $p \in (0, 1]$, and the sweep of Section 4.2 reports what the linear case owns quantitatively. One asymmetry holds across all three: bound mass already updates the size margin, $M_o = M_o(0) + r_o$ in eq. (9), but never the match margin. Size learns; match does not.

3.4 Limit allocations

The two roles that the binding law separates become exact in the two limits of the absorption tempo, and the destination result of Section 4 is the path between them.

Proposition 1 (fast absorption) *Fix the occupation set. As $\theta_{abs}/T_{shock} \to 0$ no capacity ever binds, and the final allocation of reinstated mass is*

$$R_o = \int w_o(\mathbf{r})\, S(\mathbf{r})\, d\mathbf{r}, \qquad w_o(\mathbf{r}) = \frac{e_o(\mathbf{r})^{\beta_m}}{\sum_{o'} e_{o'}(\mathbf{r})^{\beta_m}}, \tag{10}$$

where $S(\mathbf{r})$ is the cumulative surviving seed density. As $\beta_m \to \infty$ the shares concentrate on the best match and the allocation converges to strict best-match assignment.

Proof. With slack capacities the absorbed fraction is one for every occupation at every step, so each step's seeding is bound in full at the claim shares of eq. (8). Absent births the readiness fields are time-invariant, so the shares w_o are as well, and summing over steps gives $R_o = \int w_o\, S$. The second statement is the pointwise limit of the sharpened shares. □

The statement is exact at the level of the discrete law. As a self-consistency check of the discrete law against its own analytic limit, at $\theta_{\text{abs}} = 0.05$ the realised allocation matches $\int w_o\, S$ to machine precision (relative L_1 error below 10^{-15}). The check verifies the implementation, not the model.

Proposition 2 (slow absorption, general capacity scaling) *Let the capacity scale as $c_o = \kappa_t\, M_o^p$ with exponent $p \in (0, 1]$ and a common rate factor $\kappa_t \propto 1/\theta_{abs}$. As $\theta_{abs}/T_{shock} \to \infty$ the final allocation converges to*

$$R_o \;=\; \Big[M_o(0)^{1-p} + (1-p)\, u^*\Big]^{1/(1-p)} - M_o(0) \quad (p < 1), \qquad R_o \;=\; S\, \frac{M_o(0)}{M_{tot}} \quad (p = 1), \tag{11}$$

where u^* is the unique scalar with $\sum_o R_o = S$ and S the total surviving seed mass. The limit depends on the initial masses alone. As $p \to 1$ it approaches the size shares; as $p \to 0$ the uniform absolute allocation S/n.

Proof. Every claim share is strictly positive, since the readiness is an exponential, so each occupation's desired intake is positive wherever unbound mass remains, while the capacities $\kappa_t M_o^p$ shrink without bound as θ_{abs} grows, uniformly across occupations; for θ_{abs} large enough every capacity binds throughout the drain, and the slack phases at the tails of the seeding pulse and the end of the drain carry a mass share that vanishes as θ_{abs} grows, exactly as at $p = 1$, since the argument uses only that all capacities go to zero. While all capacities bind, absorption follows $\dot{M}_o = \kappa_t M_o^p$ with $M_o(t) = M_o(0) + r_o(t)$. The time change $du = \kappa_t\, dt$ makes the system autonomous, $dM_o/du = M_o^p$, which is separable: $M_o(u)^{1-p} = M_o(0)^{1-p} + (1-p)\,u$ for $p < 1$, and $M_o(u) = M_o(0)\, e^u$ at $p = 1$, a common exponential factor that preserves the proportions. Evaluating at the u^* where the cumulative absorption equals S gives the statement; R_o is continuous and strictly increasing in u^*, so u^* is unique. □

For $p < 1$ the limit allocation is strictly increasing and strictly concave in $M_o(0)$: sublinear capacity relieves small occupations, and the slow endpoint compresses from the size shares toward the uniform allocation as p falls. Generically both remain far from the match allocation of Proposition 1, since that limit depends on the readiness geometry and this one on the initial masses alone. The divergence of the two limits therefore survives every $p \in (0, 1]$; what the exponent changes is the character of the slow endpoint, not its existence. The rate factor κ_t may carry any common time dependence, so the level-neutral normalisation used in the numerical sweep of Section 4.2, which preserves the aggregate absorption rate for every p, is absorbed by the time change and leaves the limit unchanged.

While all capacities bind, the aggregate obeys $\dot{U}_{\mathrm{tot}} = \zeta_{\mathrm{tot}}(t) - M_{\mathrm{tot}}/\theta_{\mathrm{abs}}$, so after seeding ends the stock drains linearly and empties in $\theta_{\mathrm{abs}}\, S/M_{\mathrm{tot}}$ years. The bound is saturated only in the deep limit. The capacity utilisation, the mass absorbed per step over the aggregate bound $(\Delta t/\theta_{\mathrm{abs}})\, M_{\mathrm{tot}}$ read at the peak of the mismatch, rises from 0.11 at $\theta_{\mathrm{abs}} = 2$ to 0.56 at 15 and 0.99 at 120. The shortfall is the spatial locality of binding: claims are proportional to $e_o^{\beta m}$, so the mass on the seeding crescent is in practice drained by the occupations near it, and the aggregate task mass overstates the absorbing mass at intermediate tempos. This is why the mismatch hump of Section 4.1 outlives the naive aggregate prediction.

3.5 A two-occupation example

The crossover between the limits is exhibited by the sweep of Section 4.2; this subsection derives it in the smallest economy that can host the race. Two occupations compete for a seeding pulse. Occupation 1 is small and well matched, with claim share $w_1 > \frac{1}{2}$ and initial mass $M_1(0)$; occupation 2 is large and poorly matched, with claim share $1-w_1$, and large enough that its capacity never binds, so the cascade is one step. The seed arrives as a rectangular pulse of total mass S over a window T. The pulse must be a flow: an instantaneous pulse saturates every capacity and match never enters, so the window is what makes the race exist. Write $\sigma_s = S/T$ for the flow, $\vartheta = \theta_{\text{abs}}/T$ for the tempo ratio, and $\bar{m} = M_1(0)/S$ for the small occupation's mass relative to the shock.

In the continuous limit of the binding law the unbound stock vanishes. While occupation 1 is unsaturated both absorb their claims; while it is saturated, the per-step residue is re-offered and taken by occupation 2 within the step, so the stock is of order the step and the example isolates the destination race, not the mismatch hump, which lives on the depth of the cascade and the spatial locality of claims that a two-occupation economy collapses. Saturation is decided at the start and can only end, never begin: absorbed mass grows the capacity $c_1 = (M_1(0)+r_1)/\theta_{\text{abs}}$ while the claimed flow $w_1\sigma_s$ is constant. Occupation 1 is saturated at the start if and only if $w_1\sigma_s > M_1(0)/\theta_{\text{abs}}$, that is $\vartheta > \bar{m}/w_1$.

Proposition 3 (two-occupation split) *The final allocation of the small occupation is*

$$\frac{r_1(\vartheta)}{S} = \begin{cases} w_1, & \vartheta \le \bar{m}/w_1, \\ w_1(1+\vartheta) - \bar{m} - w_1\vartheta\,\ln\dfrac{w_1\vartheta}{\bar{m}}, & \vartheta > \bar{m}/w_1, \quad \vartheta\ln\dfrac{w_1\vartheta}{\bar{m}} < 1, \\ \bar{m}\left(e^{1/\vartheta}-1\right), & \vartheta\ln\dfrac{w_1\vartheta}{\bar{m}} \ge 1, \end{cases} \tag{12}$$

and $r_2 = S - r_1$. The split is continuous, constant below $\vartheta^* = \bar{m}/w_1$, and strictly decreasing above it. It recovers the match allocation $r_1 = w_1 S$ as $\vartheta \to 0$ and the saturated allocation, $r_1 \to 0$ with the full pulse cascading to the large occupation, as $\vartheta \to \infty$. The crossover is at $\vartheta^* = M_1(0)/(w_1 S)$: the small occupation's mass over its claim on the shock.

Proof. Below ϑ^* occupation 1 is unsaturated at the start and, since c_1 grows while $w_1\sigma_s$ is constant, stays so; both occupations absorb their claims throughout the pulse and $r_1 = w_1 S$. Above ϑ^* occupation 1 is saturated from the start and absorbs at capacity, $\dot{r}_1 = (M_1(0)+r_1)/\theta_{\text{abs}}$, so $r_1(t) = M_1(0)\,(e^{t/\theta_{\text{abs}}}-1)$ while saturation lasts; occupation 2 takes the residue $\sigma_s - c_1(t)$. De-saturation occurs when the grown capacity meets the claimed flow, $c_1(t_d) = w_1\sigma_s$, at $t_d = \theta_{\text{abs}}\ln(w_1\vartheta/\bar{m})$. If $t_d < T$, that is $\vartheta\ln(w_1\vartheta/\bar{m}) < 1$, occupation 1 absorbs its claim $w_1\sigma_s$ for the remaining $T - t_d$, giving the middle branch; if $t_d \ge T$ saturation lasts the whole pulse, giving the third. The branches agree at both boundaries. Within the middle branch $d(r_1/S)/d\vartheta = -\,w_1\ln(w_1\vartheta/\bar{m}) < 0$, and within the third $d(r_1/S)/d\vartheta = -\,\bar{m}\,e^{1/\vartheta}/\vartheta^2 < 0$, so the split is strictly decreasing above ϑ^* and monotone throughout. □

With $w_1 > \frac{1}{2}$ the tempo reverses the majority: below ϑ^* the well-matched specialist takes most of the reinstated work, and far above it the large, poorly matched occupation takes essentially all of it, the forced downgrading in miniature. The saturated phase is the $p = 1$ growth of Proposition 2; under the sublinear law $c_1 = M_1^p/\theta_{\text{abs}}$ the same phase integrates to M_1^{1-p} linear in time, the closed form of eq. (11). Figure 5 is the many-occupation version of this mechanism, with the single crossover ϑ^* smeared by the distribution of sizes and claims.

4 Results

We run a single technology field, calibrated to task-level AI exposure as in Storck (2026), through the dynamic economy. The field matures over $T_{\text{shock}} = 5$ years; we vary the redistribution tempo against this fixed shock, writing θ for the pooled setting $\theta_L = \theta_{\text{abs}} = \theta$. The two timescales are separated only in the decomposition check of Section 4.2. The economy parameters are the static model's, restated in Section 2; the dynamic layer adds the redistribution timescales θ_L and θ_{abs} (both 3 years at reference), the claim sharpness $\beta_m = 3$, the diffusion window $T_{\text{shock}} = 5$ years, and the step $\Delta t = 0.2$ years. Occupational birth is switched off in the runs of this section, so the occupation set is identical across regimes and the allocation comparison is well defined. In every run the population sum is conserved to numerical precision and the unbound stock drains below 10^{-8}; both are asserted by the producing scripts.

4.1 The transitional mismatch

Figure 3 shows the shock and the mismatch it creates. The left panel is the logistic diffusion of (7), rising over the five-year window. The right panel is the unbound stock $U(t)$ for four redistribution tempos, $\theta \in \{1, 3, 8, 15\}$ years. When redistribution is faster than the shock, the mismatch is barely visible, since the market binds the displaced work as fast as it is seeded. As redistribution slows, the mismatch hump grows by a factor of eighty from the fastest to the slowest tempo shown, and peaks later, moving from $t \approx 2.6$ to $t \approx 3.6$ years within the adoption window. In every case it drains to zero once diffusion saturates and binding catches up. The shape is that of the mismatch index Şahin et al. (2014) measure over the Great Recession, which likewise rises and drains with no structural shift. The driver here is task destruction and creation rather than the business cycle. There is no permanent mismatch in the model: all created work is eventually bound. The transitional cost lives in the height and persistence of the hump, not in the endpoint, and the static model cannot register it.

4.2 The destination depends on the tempo

The hump measures the cost of the transition; the same ratio also determines its outcome. Figure 4 shows where the reinstated work is bound in two regimes, run through the same

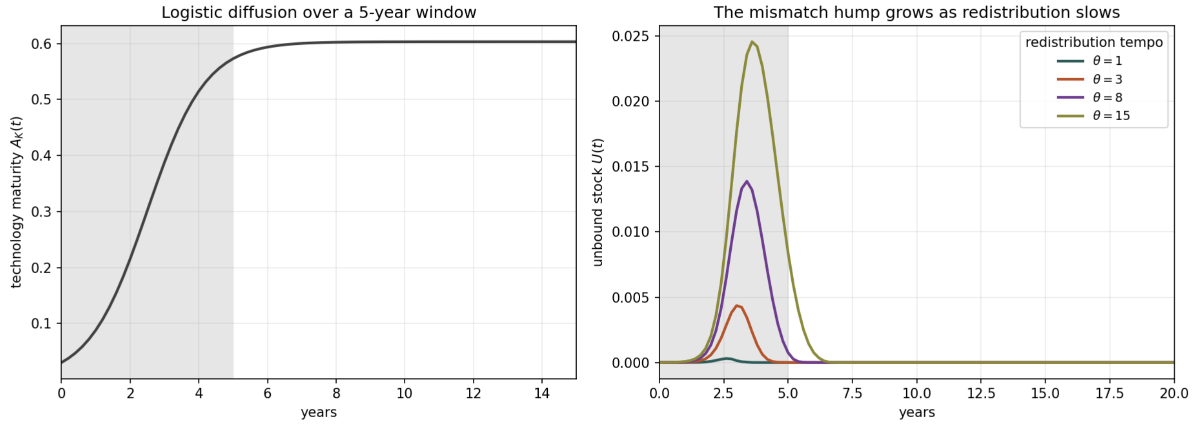

Figure 3: Calendar-time dynamics. Left: the technology matures as a logistic diffusion over a five-year window. Right: the unbound stock $U(t)$, the transitional mismatch between destruction and creation, for four redistribution tempos $\theta \in \{1, 3, 8, 15\}$ years against the fixed five-year shock. The mismatch hump grows and lingers as redistribution slows relative to the shock, but always drains to zero. The transitional cost is in the path rather than the endpoint.

technology and the same field. In the gradual regime ($\theta = 1$ year against the five-year shock), the best-matched occupations keep pace and absorb the work. The gainers are the high-skill specialists whose capabilities sit at the frontier of the field. In the congested regime ($\theta = 15$ years), the small best-matched occupations cannot bind fast enough. Their size-limited capacity $c_o = M_o/\theta_{\text{abs}}$ is small, the mass accumulates in U, and the large, lower-skilled occupations that can absorb it quickly take it instead. The same work is downgraded. Large low-skill service occupations have played the absorbing role before, and D. H. Autor and Dorn (2013) document their growth as routine work was automated away. The leading gainers make the point concretely. In the gradual regime they are preventive-medicine physicians, physicists, and natural-sciences managers. In the congested regime they include fast-food and counter workers, waiters, and dishwashers. Table 1 ranks the top ten in each regime.

The shift is not marginal, and the mass carries the evidence. Define an occupation's unconstrained claim as the seed-weighted share the claim law (8) would allocate with no size cap, and its size as the pre-shock task mass that caps the rate. In the gradual regime the top claim quartile takes 49 percent of the reinstated mass; in the congested regime the top size quartile takes 57 percent, and the bottom size quartile is squeezed from 22 to 4 percent. The correlations summarise the same reallocation. The final allocations of reinstated mass in the two regimes correlate at +0.37 (Pearson, unweighted across the 849 pre-existing occupations, +0.31 by rank), and weighting by baseline employment raises the correlation to +0.68, because the largest occupations gain in both regimes. The divergence is concentrated in whether specialist best matches or large absorbers take the frontier work, which is the margin the result is about. One technology and one field produce

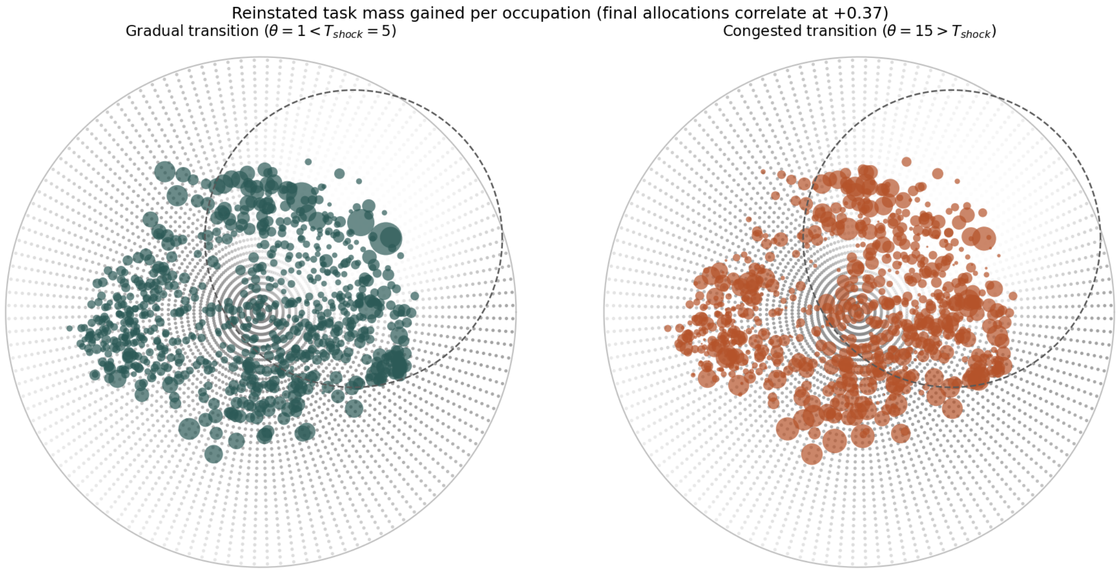

Figure 4: The destination of reinstated work depends on the tempo ratio. Both panels show the reinstated task mass gained by each occupation, over the survival-gated seeding field (grey) with the technology field dashed, for the same five-year shock. Left, a gradual transition: the best matches keep pace and high-skill specialists at the frontier absorb the work. Right, a congested transition: small best-matched occupations cannot keep up and the work is forced onto large, lower-skilled occupations. The two final allocations correlate at +0.37.

substantially different destinations, differing only in the tempo of the shock relative to the market's capacity to respond. This is the destination counterpart of the transitional hump. The fixed point sees the endpoint of a frictionless reallocation; the endpoint the economy actually reaches depends on how fast it was forced to move.

The mechanism reads off the per-occupation correlations. In the gradual regime absorption tracks the claim, with Pearson +0.83 and rank +0.92, and is uncorrelated with size (+0.04). In the congested regime the claim decouples (+0.37, rank +0.21) and size takes over (rank +0.95). All rankings are in absolute mass; the relative-change measure is unbounded for small occupations and is not used. Rank agreement with size is not equality with global size shares: seeding is spatially local, so the congested allocation is an interior point between the limits, not the slow limit of Proposition 2.

Five robustness checks bound the claim. Sweeping the sharpness β_m from 1 to 8 moves the cross-regime correlation from +0.26 to +0.61. The divergence of destinations survives the full range while its point size depends on the handle. Decomposing the pooled tempo into its two timescales shows which one binds. Slowing only absorption ($\theta_L = 1$, $\theta_{\text{abs}} = 15$) reproduces the congested allocation (correlation +0.39 with the gradual run), while slowing only mobility ($\theta_L = 15$, $\theta_{\text{abs}} = 1$) leaves the destination essentially unchanged (+0.996). The forced downgrading is a statement about how fast occupations can take on new work, not about how fast workers move.

Table 1: Leading gainers of reinstated mass, top ten per regime, share of the regime's total in parentheses. The heads of the two lists overlap, which is the employment-weighted correlation of +0.68 read occupation by occupation; the reshuffling below them is the unweighted +0.37. Both lists mix skill levels; the claim is the shift in composition, not a clean partition.

	Gradual ($\theta = 1$)	Congested ($\theta = 15$)
1	Preventive medicine physicians (1.4%)	Fast food and counter workers (0.7%)
2	Physicists (1.3%)	Preventive medicine physicians (0.7%)
3	Natural sciences managers (1.0%)	Waiters and waitresses (0.7%)
4	Chief executives (0.8%)	Dishwashers (0.6%)
5	Dishwashers (0.6%)	Solar energy installation managers (0.6%)
6	Mechanical engineering technologists and technicians (0.6%)	Receptionists and information clerks (0.6%)
7	Aerospace engineering and operations technologists and technicians (0.6%)	Food preparation workers (0.6%)
8	Epidemiologists (0.6%)	Electricians (0.6%)
9	Bioengineers and biomedical engineers (0.5%)	Dining room and cafeteria attendants and bartender helpers (0.5%)
10	Mechanical engineers (0.5%)	Medical and health services managers (0.5%)

The third check varies the survival gate itself, since the low-priced character of surviving seed is inherited from it. Both regimes are rerun with the gate off, and with a comparative-advantage gate, a human-productivity field $h(\mathbf{r}) = \exp[\lambda_h(1 - \hat{\phi}_K(\mathbf{r}))]$ with $\hat{\phi}_K$ the unit-amplitude technology shape, so labour is more productive where the machine is weak and capital operates where $s_K\phi_K > hR/\Pi$. The sweep takes λ_h up to 2, and $\lambda_h = 0$ recovers the uniform-R gate. The divergence is intact in every variant. The cross-regime correlation stays between +0.35 and +0.46, and the congested size rank correlation between +0.91 and +0.96. What the gate moves is the gainers' price level rather than the tempo dependence. Without the gate the capital-dominated core seeds, and the congested absorbers become large health and administrative occupations rather than food service. The comparative-advantage gate raises the mass-weighted gainer price with λ_h. The downgrading, congested gainers priced below gradual gainers, persists throughout.

The fourth check generalises the capacity law itself, since linear scaling $c_o = M_o/\theta_{\text{abs}}$ is one of the structural assumptions of Section 3.3. A level-neutral sublinear law, $c_o \propto M_o^p$ with the aggregate absorption rate preserved so that the exponent moves only the cross-sectional allocation of capacity and not the effective tempo, is run at $p \in \{0.5, 0.7\}$ against the linear baseline. The theorem survives and the point magnitude does not. At the deep endpoint the realised allocation approaches the closed form of Proposition 2, relative L_1 of 0.11 and 0.16 against 0.28 for the size shares at $p = 1$, and stays uncorrelated with the fast endpoint (+0.14 and below), so the divergence of the limits holds at every exponent. At the reference tempos the cross-regime correlation rises from +0.37 to +0.53 at $p = 0.7$

and $+0.68$ at $p = 0.5$, the congested size-rank correlation falls from $+0.95$ to $+0.49$, and the top size-quartile share from 57 to 35 per cent: sublinear capacity relieves small occupations, the crossover moves out, and the downgrading at a given tempo weakens. The path-dependence of the destination is not a property of the linear law; the severity of the downgrading at a given tempo is.

The fifth check releases the second structural assumption, frozen readiness. In the update variant, bound mass updates the occupation's bundle by mass-as-relevance renormalisation, $b'_o = (M_o(0)\, b_o + \rho_o)/(M_o(0) + r_o)$ with ρ_o the cumulative binding density, and the centroid, the capability levels, and the readiness re-integrate over the grown bundle each step; the wage-bearing bundle is untouched, since displacement takes the payment, not the capability. Occupations then move toward the work they take, and the movers are the gradual winners: the employment-weighted mean centroid drift is well below the readiness scale ℓ in both regimes (0.012 and 0.024 against $\ell = 0.133$), but weighted by reinstated mass the gradual regime drifts 0.12, comparable to ℓ, and the drift follows the bound share $r_o/(M_o(0) + r_o)$ (rank $+0.91$ in the gradual regime, $+0.55$ in the congested, where capacity rather than match governs what binds). The congested regime is nearly unaffected, with the allocation displaced by 3 per cent and its size rank intact at $+0.94$, while the gradual allocation displaces by 22 per cent toward the well matched. The cross-regime correlation falls from $+0.37$ to $+0.09$: updating widens the divergence. Frozen readiness is therefore the conservative choice, and the frozen headline numbers understate the path-dependence that learning from absorbed work would produce.

4.3 The homotopy between the limits

The tempo result is the path between the two proved endpoints of Section 3.4. Figure 5 traces the final allocation across θ_{abs} from 0.5 to 120 years against the fixed five-year shock, with mobility held fast at $\theta_L = 1$, since the decomposition above shows the destination is carried by θ_{abs}. Its correlation with the fast-limit allocation of Proposition 1 falls monotonically from $+1.00$ to $+0.07$, and its correlation with the size shares, the slow limit of Proposition 2 at $p = 1$, rises monotonically from $+0.03$ to $+0.94$ ($+0.996$ by rank) over the same sweep. The crossover, where the two associations are equal, sits between $\theta_{\text{abs}} = 10$ and 15, two to three shock windows. The simulation's remaining role is to locate the crossover, since the two endpoint allocations are derived.

5 The general-equilibrium price feedback

Both the static regime and the dynamic transition above hold the price field $\Pi(r)$ fixed. The shock relocates labour and re-sorts it, congestion adjusts the marginal value of crowded destinations, but the per-unit price of capability content does not re-solve. This is a short-

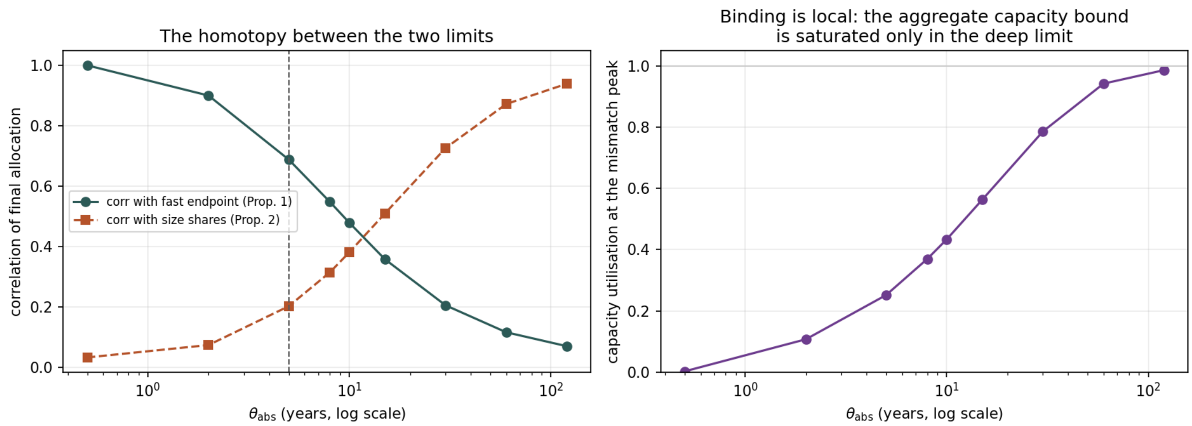

Figure 5: The destination as a homotopy between proved limits. Left: the correlation of the final allocation with the fast-limit claim allocation (Proposition 1) and with the slow-limit size shares (Proposition 2, $p = 1$), across the absorption tempo at fixed mobility; the vertical line marks the shock window. Right: the capacity utilisation at the mismatch peak, the mass absorbed per step over the aggregate bound. The shortfall at intermediate tempos is the spatial locality of binding, and Proposition 2 is the regime where it reaches one.

run reading. The price field is an equilibrium object, derived as a capability assignment in Storck (2026). Its level Π_0 is the estimated field of eq. (2); holding it fixed discards part of the model's own content. Acemoglu and Restrepo (2026) make the point sharply. In their task model the base wage *adjusts* to automation through their propagation matrix Θ, and that adjustment is the mechanism. Our fixed-price layers hold both the rent wedge and the base price fixed, so to "derive the prices" in their sense we must let the base price move. This section closes that loop.

The closure: level from data, change from the assignment. We keep the level from the estimated field, the empirical anchor, and take the *change* from the assignment, exactly as Acemoglu and Restrepo (2026) take the wage level as given and compute its response. Their wage law $w_g \propto (\Gamma_g/\ell_g)^{1/\lambda}$ ties the wage to tasks per worker, and its spatial form is

$$\Delta \ln \Pi(r) \;=\; \frac{1}{\sigma}\Big[\,\ln\big(1-a(r)\big) \;-\; \ln\frac{n_L(r)}{n_L^0(r)}\,\Big], \tag{13}$$

applied to the baseline, $\Pi(r) = \Pi_0(r)\,e^{\Delta \ln \Pi(r)}$. Automation is a *demand* shock to labour. Where capital takes the task (a high), the task leaves the human set and the wage falls, and displaced labour crowding the remaining work ($n_L > n_L^0$) lowers it further. Both terms are stabilising. The sign is not innocuous. The opposite reading, automation as a *supply* shock that makes scarce human work dearer, inverts equation (13) and is destabilising, since a dearer wage opens the takeover gate. Run as a single pass at $\sigma = 3$, the scarcity reading collapses the labour share to 0.33 (capital chases the work it has just made dear), while the demand reading lifts it to 0.83. The fixed-price value 0.58 sits between. Labour shares in this section hold the allocation at its baseline, which isolates the price channel. The headline shares of Storck (2026) re-solve the allocation as well and are correspondingly higher, 0.626 at fixed prices and 0.78 at the fixed point below. The headline depends on the sign, which therefore must be chosen on economic grounds, and it stands on the model's own ground before any external authority. The estimated field is an assignment price rather than a scarcity price: the socio-cognitive north is both highly priced and densely staffed, so scarcity does not generate the level (Storck 2026), and a change priced by scarcity would be priced by a mechanism the level itself rejects. Inside the closure, the demand reading is the self-correcting one, and self-correction is what lets a fixed point exist at all; the scarcity collapse above is the self-amplification of the alternative. Nor is the choice delicate in the elasticity: across σ from 0.5 to 5 the fixed-point lift of the labour share stays positive throughout and the log price level scales as $1/\sigma$, from -1.30 to -0.17, so the sign-dependent quantities change magnitude but never direction. Acemoglu and Restrepo (2026) corroborate the choice from outside the model, since automation dissipates rents and lowers wages.

The static fixed point. Iterating price and allocation to mutual consistency converges, and the convergence has an interpretable dominant term and a measured certificate. Differentiating the gate (3) gives $\partial \ln(1-a)/\partial \ln \Pi = -a\,(R/\Pi)/\tau$: a lower price closes the gate, and the closing gate raises the price back. The gate contribution to the slope of the map (13),

$$K_{\text{gate}} \;=\; \frac{1}{\sigma}\,\sup_{\mathbf{r}}\,\frac{a(\mathbf{r})\,R}{\Pi(\mathbf{r})\,\tau}, \tag{14}$$

peaks at 1.64 at the converged point. The crowding response of the re-sorted density contributes a second term that is not signed by inspection, and we do not derive its operator bound; measured, it lowers the slope, since the spectral radius of the full map at the fixed point, crowding included, is 1.59 by power iteration, below the gate supremum. Damped iteration with weight d converges locally when the slope is stabilising and satisfies $K < 2/d - 1$, which at $d = \frac{1}{2}$ is 3, a factor-two margin; the power iteration returns a modulus, and the evidence that the eigenvalue is of the stabilising sign is the observed convergence itself, from every start below. The certificate is local, a derivative at the point, and global uniqueness is not claimed. As evidence beyond the point, the iteration lands on the same fixed point from four starting fields, zeros, both one-pass extremes including the wrong-signed scarcity field, and a seeded random perturbation, and at every converging damping $(\frac{1}{4}, \frac{3}{8}, \frac{1}{2})$, with the labour share agreeing to 3×10^{-5}. The slope scales as $1/\sigma$ ($\sigma K = 4.77$ at $\sigma = 0.5$ against 4.78 at $\sigma = 3$), which places the failure boundary at $d = \frac{1}{2}$ near $\sigma = 1.6$. At $\sigma = 1$ the iteration indeed fails at $d = \frac{1}{2}$ and converges at $d = \frac{1}{4}$ and $\frac{1}{8}$ to the same point. The fixed point does not depend on the damping; only the path to it does, and the baseline iteration converges in nine damped steps.

The self-consistent state is an interior point rather than the one-pass extreme. The labour share settles at 0.73 (against 0.58 at fixed prices), the log price level falls 0.26, and displacement is cut by roughly 44 percent (D_o from 0.28 to 0.16). The wage adjustment absorbs most of the shock. Human work becomes about a quarter cheaper, capital adopts far less, and the labour share stays high, but the harm now sits in the wage *level* rather than in employment, the rent- and wage-dissipation of Acemoglu and Restrepo (2026).

The dynamic close, and statics meeting dynamics. Wired into the transition, with the price responding to the current adoption and allocation at every step through equation (13) with a one-step lag, the feedback shapes the whole path. The wage level falls monotonically as adoption rises, from -0.09 early to a plateau at -0.29, and the displacement wave is markedly gentler. The unbound-mass peak falls by about 63 percent (from 0.44 to 0.16 percent of the workforce) at the same timing, and the reinstated stock ends lower, because the falling wage deters the automation that would have produced the mismatch. The transition lands where the static fixed point says it should. The dynamic end-state labour share (0.739) matches the static fixed point (0.73); the log price levels (-0.285 against -0.26) differ by about ten per cent of the value. Two independent constructions, static fixed-point iteration and dynamic time-stepping, land within this gap of the same equilibrium, so the price feedback is a property of the model rather than of either method, and the static and dynamic accounts cohere.

What this does and does not establish. The feedback is first-order. At fixed prices both layers overstate the displacement and understate the labour share, and they overstate the violence of the transition. The stabilising wage adjustment is the shock absorber, and it operates throughout the transition. The quantitative point estimate depends on the assignment elasticity σ and on the reduced-form closure, since equation (13) is the AR wage law applied region-wise rather than a full re-solve of the assignment programme of Storck (2026). The AI field remains one illustrative instance of a technology field rather than a measurement. Across σ from 5 to 0.5 the fixed-point lift of the labour share runs from $+0.11$ to a peak of $+0.20$ at $\sigma = 1$, and the damping of the mismatch hump rises from 46 to 89 per cent. We therefore read the result as establishing the *direction and order of magnitude* of the price feedback, and the mutual consistency of the static and dynamic closes, rather than a calibrated forecast of the labour share.

6 Discussion

The result is that the tempo of a technology shock, relative to the speed at which the labour market reallocates, determines where the reinstated work ends up. A gradual diffusion lets work find the occupations whose capabilities match it; a shock that outruns adjustment forces a downgrading that persists in the final allocation. The same field and the same seeded reinstatement yield different allocations of work depending on the pace. The task-based equilibrium framework is not built to make this statement. Its rest point is the destination of a frictionless economy, and the friction here is time itself.

Two qualifications bound the claim, the first inherited and the second introduced by the dynamic layer. The first concerns the survival gate. The static model places surviving reinstatement on the low-priced side of the boundary, because a seeded task is retained by labour where labour is the cheaper producer at a *uniform* cost R. In Acemoglu and Restrepo (2019) reinstatement is instead new *complex* work in which labour has a comparative advantage, which requires the survival condition to turn on relative rather than absolute productivity. Section 4.2 runs exactly this variant, alongside the gate removed altogether, and the gate turns out to govern the *character* of the destination rather than its tempo dependence. The divergence and the size mechanism survive every variant while the gainers' price level moves with the gate. What remains conditional on the survival-gate economics is therefore the high- versus low-skill reading of the congested absorbers, food service under the absolute gate and large health and administrative occupations without it, rather than the result that a fast shock forces the work onto whoever is large enough to take it.

The second qualification is the binding calibration. The allocation depends on the sharpness β_m and the absorption timescale θ_{abs}, and Section 4.2 reports both sensitivities. The divergence of destinations survives the sharpness sweep with a point size that moves with the handle, and the absorption timescale alone carries it. The remaining quantitative claim is the regime boundary, how slow absorption must be relative to the shock for downgrading to set in. Section 3.5 derives it in the minimal economy, $\vartheta^* = M_1(0)/(w_1 S)$ in the tempo ratio, and the tempo sweep exhibits it in the full one, with the single crossover smeared by the distribution of sizes and claims.

Several limitations remain. The realisation is illustrative, inherits the calibration and discipline of Storck (2026), and supports no causal claim. The unbound stock drains to zero, so the model has no permanent mismatch and no involuntary non-employment. The transitional cost is the hump, and it is a model-internal cost: a model with task decay or a residue that never binds would carry the shock speed as permanent non-employment or earnings loss, while here the permanent imprint is confined to the allocation of the reinstated work. The lasting individual imprint is empirically real and uneven, since Edin et al. (2023) find moderate mean earnings losses from occupational decline, concentrated on low earners. That extension is left for later work. New tasks enter as added weight on existing locations rather than as new points on the disk; this is the model's micro-task ontology rather than a shortcut, since a location holds a continuum of micro-tasks and the discrete task statements are the measurement layer's aggregation, with condensation of bound mass into new task points deferred to the birth machinery. And the destination results are read in absolute and spatial terms, for the reason stated in Section 4.2.

The downgrading of the congested regime has an empirical silhouette. Beaudry, Green, and Sand (2016) document a cascade after 2000. As demand for cognitive tasks fell, high-educated workers moved down the occupational ladder and pushed the less educated further down and out. The direction is reversed here. In their account workers descend the ladder because demand for their tasks has fallen; in the congested regime the work itself descends the match ranking because the occupations that fit it best are too small to absorb it in time. The same outward pattern, work performed below the skill level that matches it, arises from different mechanisms, one demand-side and one absorption-side, and the cascading image is theirs.

Measurement and testable implications. The absorption timescale is the model's one load-bearing free quantity, so the theory should state what would measure it and what would test it. Observationally, θ_{abs} is the time an occupation needs to absorb its own size in new task content: its empirical counterparts are the revision rate of occupations' task lists, the uptake of new job titles within occupations (Lin 2011; D. Autor et al. 2024), and text-based measures of within-title task turnover, which find substantial task content change inside narrowly defined titles over decades (Atalay et al. 2020). The order of magnitude has a yardstick. The crossover of Section 4.3 sits at two to three shock windows, so for a five-year diffusion the downgrading regime begins where occupations need a decade or more to absorb their own size in new content. Against measured diffusion speeds, the decades of David (1990) and the adoption lags of Comin and Hobijn (2010), most historical technologies were gradual by this yardstick; the question the theory poses, and does not answer, is whether current diffusions are the first to cross it.

Three implications are testable in principle. Faster diffusion episodes place reinstated work in larger, lower-paid occupations, and the gainer composition of Table 1 is the cross-episode prediction. The transitional mismatch scales with the tempo ratio rather than with the size of the shock. And the operative friction is absorptive capacity, not worker mobility, so episodes that differ in mobility at similar absorption speed should share destinations, the decomposition of Section 4.2 run on data. Calibrating θ_{abs} against these measures is left to empirical work.

7 Conclusion

Turning the spatial task-based model from an equilibrium into laws of motion requires one new object, a stock of unbound task mass, and yields two things the equilibrium cannot. The first is the transitional mismatch $U(t)$, the gap between destruction and creation as a transient that grows with the tempo of the shock and drains as the market catches up. The second, and the central result, is that the tempo of the shock relative to the speed of redistribution determines the destination of the reinstated work: a gradual shock matches it to well-suited specialists, a congested one forces it onto large lower-skilled occupations. Both endpoints are proved as limits of the binding law, for every capacity-scaling exponent, the crossover between them is derived in closed form in a two-occupation economy, and the realised destination moves monotonically between the limits with the tempo. The general-equilibrium close leaves both results standing: the price feedback is stabilising, and the static and dynamic accounts land on the same fixed point. The same technology yields a different distribution of work depending only on how fast it arrives. The cost and the outcome of automation are, in this sense, properties of the path and not only of the endpoint.

A The inherited economy

The dynamics consume the static economy across one interface, and this appendix restates everything that crosses it, so that every parameter of the model traces to a stated source within this paper. The construction and validation of the geometry remain with Storck and Andersson (2026); the model of technological change and its calibration with Storck (2026). Table values reproduced here are adapted from those papers. Table 3 consolidates the parameters; the sections below state where each number comes from.

Geometry

The task space is built from the 17,606 task statements of O*NET, each encoded by a text-embedding model as a vector in a high-dimensional semantic space. The embedding matrix is centred and projected on its first two principal components, and the scores are read in polar coordinates: the angle ξ is a domain orientation, the radius $\chi \in [0,1]$ a normalised specificity. The orientation of the map is frozen against a fixed reference, so runs, encoders, and database releases are comparable. Occupations enter as importance-weighted bundles $b_o(\mathbf{r})$ of their task points with centroid μ_o; 849 occupations with observed employment enter the dynamic economy, weighted by their OEWS employment shares L_0. The choice of encoder, the recovery of the expert-curated job-family classification, and the external validity of the coordinates are established in Storck and Andersson (2026) and not reproduced here.

The price field

The price of skill is the reduced form of eq. (1), estimated once on the occupational wage cross-section (785 occupations with observed mean wages) and held fixed under the shock. The coefficients and their HC3 standard errors are those of eq. (2), with $R^2 = 0.523$. Two structural checks discipline the form. The isotropic depth term is indistinguishable from

Table 2: Capability plane coefficients, $q_k(\mathbf{r}) = \bar{v}_k + \chi\,(u_{1,k}\cos\xi + u_{2,k}\sin\xi)$. R^2 is the plane fit; ΔR^2 the gain from level harmonics and isotropic depth. Only $S1$ and $S2$ are priced. Adapted from Storck (2026).

Cluster	$\bar{v}_k$	$u_{1,k}$	$u_{2,k}$	Pole	R^2	ΔR^2
S1	2.68	1.09	1.42	52°	0.74	0.001
S2	1.37	−2.22	1.23	151°	0.69	0.008
A1	3.20	0.95	1.31	54°	0.75	0.000
A2	1.57	−1.54	−0.24	189°	0.57	0.009

zero ($m_3 = -0.065$, $p = 0.47$): depth pays only through its interaction with direction. A second-harmonic test balanced with level harmonics does not reject the single first harmonic of the depth return ($p = 0.44$); the remaining higher-order structure is a level effect, not additional depth pricing. The step from centroid wages to task-location prices is validated by bundle integration: pricing an occupation as $\int \Pi\, b_o$ reproduces observed wages about as well as evaluating the field at the centroid (R^2 0.51 against 0.52), which licenses operating the field at task locations.

Readiness

The readiness primitive of eq. (4) needs the requirement fields q_k, the weights v_k, and three scalars. Each capability cluster's requirement field is a plane over the disk, $q_k(\mathbf{r}) = \bar{v}_k + \chi\,(u_{1,k}\cos\xi + u_{2,k}\sin\xi)$; Table 2 restates the coefficients. The gain from adding level harmonics and isotropic depth to the planes is at most $\Delta R^2 = 0.009$, so the plane is not a restrictive form. The weights v_k are wage returns from a replication of the mediation regression of Storck and Andersson (2026) on this sample: $v_{S1} = 0.328$ (0.065), $v_{S2} = 0.049$ (0.021). The ability clusters carry negative or negligible estimates and are excluded by the sign restriction $v_k \geq 0$, so the priced set is $K = \{S1, S2\}$. The acquisition scale $\ell = 0.133$ is one within-direction standard deviation of the priced-capability index, the deficit at which readiness falls to e^{-1}. The locality $\rho = 0.5$ and the over-qualification weight $\lambda_{\text{over}} = 1$ are set, not estimated.

The technology field

A technology is the Gaussian field of Section 2 with centre $\mathbf{p}_K$, reach z_K, and mature amplitude $\bar{A}_K$. The field run in Section 4 is calibrated to task-level AI exposure: each task statement carries the exposure score $\beta_E = E1 + \frac{1}{2}E2$ of Eloundou et al. (2024), the scores are smoothed to a between-cell exposure surface on the disk, and the Gaussian is fitted to that surface by least squares. The fit places the centre at $(\xi_K, \chi_K) = (38.4°, 0.463)$ with reach $z_K = 0.583$ and mature amplitude $\bar{A}_K = 0.603$, and explains $R^2 = 0.82$ of the smoothable between-cell surface (0.25 of raw task-level exposure). The projection is

Table 3: Parameter provenance. The wage wedge ω_g and the texture ledger of Storck (2026) do not cross the interface and are not used by the dynamics.

Parameter	Value	Provenance
$m_0, \ldots, m_5$	eq. (2)	estimated: wage cross-section, $N = 785$, HC3, $R^2 = 0.523$
$v_{S1},\ v_{S2}$	0.328, 0.049	estimated: mediation-regression replication, s.e. 0.065, 0.021; $v_k \geq 0$ excludes A1, A2
$\bar{v}_k, u_{1,k}, u_{2,k}$	Table 2	estimated: capability plane fits
ℓ	0.133	rule: one within-direction SD of the priced-capability index gives readiness e^{-1}
$\rho,\ \lambda_{\text{over}}$	0.5, 1	set: attachment locality and over-qualification weight
R	18	calibrated: the gate binds in the calibrated core, capital cost ≈ 30 at the centre against $\Pi \approx 47$
τ	0.08	set: width of the adoption margin, eq. (3)
β	0.5	normalised: congestion benchmark
γ	0.5	set: seeded fraction of displaced mass; swept in the static companion
ν	11.8	rule: one SD of baseline occupation value per logit unit, evaluated at $A_K = 0$
c_m	23.0	rule: the median move costs one ν
$\xi_K,\ \chi_K$	38.4°, 0.463	calibrated: least-squares fit to the exposure surface
z_K	0.583	calibrated: same fit
$\bar{A}_K$	0.603	calibrated: same fit
s_K	1	normalised
$\theta_L,\ \theta_{\text{abs}}$	3 yr	reference; swept in Section 4
β_m	3	reference claim sharpness; swept 1–8
T_{shock}	5 yr	normalisation of the time axis
Δt	0.2 yr	numerical step
σ	3	reference assignment elasticity; swept 0.5–5

one instance of a technology field, an illustration rather than a measurement; nothing in the dynamics depends on the field being AI. The scale s_K is normalised to one, and the maturation $A_K(t)$ rises from zero to $\bar{A}_K$ along the logistic of eq. (7).

Parameter provenance

Table 3 consolidates the economy. Estimated parameters carry their regression source and standard error; calibrated parameters their target; rule-derived parameters their rule; the rest are normalised, set, or swept, as stated.

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